Solutions, Sixth Annual ECC Undergraduate Mathematics Competition, March 29, 2003

1. Right triangles.

Let $a = 14 \cdot 10^{n+1}$, $b = m - 2$ and $c = m + 2$, where $m = 245 \cdot 10^{2n+1}$. Then

$$\begin{aligned} c^2 - b^2 &= (m^2 + 4m + 4) - (m^2 - 4m + 4) \\ &= 8m \\ &= 8 \cdot 245 \cdot 10^{2n+1} \\ &= 1960 \cdot 10^{2n+1} \\ &= 14^2 \cdot 10^{2n+2} \\ &= a^2, \end{aligned}$$

which shows that the angle opposite side c is a right angle.

2. Probability of solving.

The probability is $\frac{19}{27}$.

The complement of the event that at least one solves the problem is that all three fail. Each has probability $\frac{2}{3}$ of failing, so by independence the probability that all three fail is $(\frac{2}{3})^3 = \frac{8}{27}$. Then the probability that at least one solves it is

$$1 - \frac{8}{27} = \frac{19}{27}.$$

3. A differential equation.

We claim $f''(-3) = 20$. Differentiating the given equation, we get

$$f''(x) = xf'(x) + f(x) = x^2 f(x) + f(x) = (x^2 + 1)f(x),$$

so $f''(-3) = 10f(-3) = 20$. ∎

4. Maclaurin series.

We show that $a_{11} = \frac{-1}{(11)(12!)}$.

From the Maclaurin series

$$\cos t = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!}$$

we have

$$\frac{1-\cos t}{t^2} = \frac{1}{2!} - \frac{t^2}{4!} + \frac{t^4}{6!} - \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k+2)!}$$

for all real t. Such power series may be integrated term by term within the interval of convergence, so

$$\begin{aligned} f(x) = \int_0^x \frac{1-\cos t}{t^2} dt &= \sum_{k=0}^{\infty} \int_0^x \frac{(-1)^k t^{2k}}{(2k+2)!} dt \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)(2k+2)!}\Big|_0^x \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)(2k+2)!}. \end{aligned}$$

By the uniqueness theorem for power series representations, this is the Maclaurin series for $f(x)$, and the coefficient of x^{11} is

$$a_{11} = \frac{(-1)^5}{(11)(12!)}.$$

5. Solve for $\frac{x}{y}$.

The only solution is $\frac{x}{y} = 1$. We have $\ln(3x-y)^2 = \ln 4xy$, so $(3x-y)^2 = 4xy$;

$$0 = 9x^2 - 10xy + y^2 = (9x-y)(x-y).$$

Thus $y = 9x$ or $y = x$. But if $y = 9x$, then either $3x - y = -6x < 0$ or $2x \leq 0$, and in each of these cases one of the terms in the original equation is undefined. Thus the only possibility is $y = x$, and $\frac{x}{y} = 1$. (We know $y \neq 0$ because otherwise $\ln(2y)$ would be undefined.)

6. An inequality.

The solution set is $(-1, 1-\sqrt{2}) \cup (1+\sqrt{2}, \infty)$.
The first inequality, $x < 2x+1$ is equivalent to $x > -1$. As for the second,

$$\begin{aligned} 2x+1 < x^2 &\iff (x-1)^2 > 2 \\ &\iff x > \sqrt{2}+1 \quad \text{or } x < 1-\sqrt{2}. \end{aligned}$$

Both hold when $x > \sqrt{2}+1$, and also when $-1 < x < 1-\sqrt{2}$. Thus the solution set is $(-1, 1-\sqrt{2}) \cup (1+\sqrt{2}, \infty)$.

7. Find the limit.

(a) The limit is 5. We have

$$\begin{aligned} \ln f(x) &= \frac{1}{x} \ln \frac{3^x + 5^x}{2} \\ &= \frac{1}{x}[\ln(3^x + 5^x) - \ln 2] \\ &= \frac{1}{x}\left[\ln 5^x \left(1 + \left(\frac{3}{5}\right)^x\right) - \ln 2\right] \\ &= \frac{1}{x}\left[x \ln 5 + \ln\left(1 + \left(\frac{3}{5}\right)^x\right) - \ln 2\right] \\ &= \ln 5 + \frac{1}{x}\left[\ln\left(1 + \left(\frac{3}{5}\right)^x\right) - \ln 2\right]. \end{aligned}$$

As $x \to \infty$, $\left(\frac{3}{5}\right)^x \to 0$, and $\frac{1}{x} \to 0$. Thus $\lim_{x\to\infty} \ln f(x) = \ln 5$, and $\lim_{x\to\infty} f(x) = 5$.

(b) The limit is $\sqrt{15}$. The limit

$$\lim_{x\to 0} \ln f(x) = \lim_{x\to 0} \frac{\ln \frac{3^x+5^x}{2}}{x}$$

has the form $\frac{0}{0}$, and by the rule of L'Hôpital the limit is

$$\begin{aligned} \lim_{x\to 0} \frac{\frac{1}{2}(3^x \ln 3 + 5^x \ln 5)}{\frac{3^x+5^x}{2}} &= \lim_{x\to 0} \frac{3^x \ln 3 + 5^x \ln 5}{3^x + 5^x} \\ &= \frac{\ln 3 + \ln 5}{2} = \frac{1}{2} \ln 15. \end{aligned}$$

Then $\lim_{x\to 0} f(x) = \sqrt{15}$.

8. The BEDCAF dictionary.

The 525th word is EBFCAD. There are $4 \cdot 5! = 480$ words beginning with A, B, C or D, and the next 120 words begin with E. Of these, the first $4! = 24$ begin with EA and the next 24 with EB. Thus the last EAxxxx word is number 504. There are $3 \cdot 3! = 18$ words beginning with EBA, EBC or EBD, making 522 words. The next three words are EBFACD, EBFADC, EBFCAD, and thus EBFCAD is word number 525.

9. Some sum.

The sum is $401\frac{1}{17} = \frac{6818}{17}$. From the fact that

$$(1.5)^4 = \left(\frac{3}{2}\right)^4 = \frac{81}{16} = 5 + \frac{1}{16}$$

we see that for $1 \le k \le 5$ we have $1 \le \sqrt[4]{k} < 1.5$ and $f(k) = 1$. From

$$\left(\frac{5}{2}\right)^4 = \frac{625}{16} = 39 + \frac{1}{16}$$

it follows that for $6 \le k \le 39$ we have $1.5 < \sqrt[4]{k} < 2.5$, and $f(k) = 2$. Similarly,

$$\left(\frac{7}{2}\right)^4 = \frac{2401}{16} = 150 + \frac{1}{16},$$

so for $40 \le k \le 150,\ f(k) = 3$;

$$\left(\frac{9}{2}\right)^4 = \frac{6561}{16} = 410 + \frac{1}{16},$$

so for $151 \le k \le 410,\ f(k) = 4$;

$$\left(\frac{11}{2}\right)^4 = \frac{14641}{16} = 915 + \frac{1}{16},$$

so for $411 \le k \le 915,\ f(k) = 5$. For $916 \le k \le 1785,\ f(k) = 6$, and for the remaining terms $1786 \le k \le 2003,\ f(k) = 7$. Thus

$$\begin{aligned}\sum_{k+1}^{2003} \frac{1}{f(k)} &= \frac{5}{1} + \frac{34}{2} + \frac{111}{3} + \frac{260}{4} + \frac{505}{5} + \frac{870}{6} + \frac{218}{7} \\ &= 5 + 17 + 37 + 65 + 101 + 145 + 31\frac{1}{7} = 401\frac{1}{7} = \frac{2808}{7}.\end{aligned}$$

SECOND SOLUTION TO PROBLEM 9

We note that

$$\begin{aligned} f(k) = n &\iff n - \frac{1}{2} < \sqrt[4]{k} < n + \frac{1}{2} \\ &\iff \frac{2n-1}{2} < \sqrt[4]{k} < \frac{2n+1}{2} \\ &\iff \frac{(2n-1)^4}{16} < k < \frac{(2n+1)^4}{16}. \end{aligned}$$

The length of this interval is

$$\frac{(2n+1)^4 - (2n-1)^4}{16} = \frac{8(2n)^3 + 8(2n)}{16} = n(4n^2+1).$$

Because this is an integer and the endpoints of the interval are not integers, $n(4n^2+1)$ is the number of integers k for which $f(k) = n$. Thus

$$\sum_{f(k)=n} \frac{1}{f(k)} = 4n^2 + 1.$$

This gives us

$$\sum_{k=1}^{\lfloor \frac{(2N+1)^4}{16} \rfloor} \frac{1}{f(k)} = \sum_{n=1}^{N} (4n^2+1).$$

If $N = 6$ we have $\frac{(2N+1)^4}{16} = \frac{13^4}{16} = 1785 + \frac{1}{16}$, so the last full block is that with $f(k) = 6$, ending at $k = 1785$, leaving $2003 - 1785 = 218$ terms with $f(k) = 7$. Thus

$$\sum_{k=1}^{2003} \frac{1}{f(k)} = \sum_{n=1}^{6} (4n^2+1) + \frac{218}{7} = 401\frac{1}{7} = \frac{2808}{7}$$

as in the first solution.

10. How many terms?.

There are

$$\frac{(2004)(2005)}{2} = 2\,009\,010$$

terms. We'll show that in general, when $(x+y+z)^n$ is expanded and like terms collected, there are $(n+1)(n+2)/2$ terms.

The number of terms is the same as the number of expressions of the form $x^a y^b z^c$ where a, b, c are nonnegative integers with $a+b+c=n$. Note that when a and b are chosen then c is determined as well. If $a=n$, there is only one choice for b, namely $b=0$. If $a=n-1$, b may be 0 or 1. If $a=n-2$, b may be 0, 1 or 2. In general, a may be any integer from 0 to n, and when a is chosen there are $n+1-a$ choices for b which make $a+b \le n$. The resulting number of choices for (a,b) is

$$1+2+3+\cdots+n+(n+1),$$

corresponding to $a=n, n-1, n-2, \cdots, 1, 0$, respectively, for a total of $(n+1)(n+2)/2$.